Partition Logics, Their Continuum Extensions, and the Conservation Law They Lack:
Four Clauses of Contextual Conservation

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Abstract

Partition logics arise from pastings of Boolean algebras associated with partitions of a set. We show that any continuum extension obtained by pasting Boolean sub-σ-algebras of a common measurable space $(\Omega, \mathcal{F})$ remains a *concrete* logic: point evaluations furnish a full, order-determining family of two-valued states, so by the Kochen–Specker theorem the construction cannot be isomorphic to the projection lattice $P(\mathcal{H})$ for $\dim \mathcal{H} \geq 3$. Read positively, the no-go isolates what such extensions lack: a *conservation law for probability across intertwined contexts*. We decompose this law into four independent clauses—(C1) noncontextuality of value, (C2) conservation in every context, (C3) continuous transitive symmetry on pure states, (C4) rigidity of the conserved transition structure—each enforced by a classical theorem (Kochen–Specker and Gleason for (C1)–(C2) at dimension ≥ 3; Hardy's continuity axiom for (C3); Wigner's theorem and the Koecher–Vinberg classification with local tomography for (C4)), and shown to be logically independent by a scorecard of known theories, each obtained by deleting exactly one clause. The dimension-two caveat is sharpened accordingly: the qubit projection lattice is a horizontal sum, hence itself concrete, so at $d = 2$ the logical clauses do not bind and the entire quantum content of the Bloch sphere—including the complementarity equality $D^2 + V^2 = 1$ and the Brukner–Zeilinger invariant—is carried by the geometric clauses alone. A continuum limit of partition logics over a classical point space therefore cannot approach quantum structure: the continuation of partition logics is a change of category, not a limit.

1 Introduction

Partition logics arise in the study of empirical propositional structures, finite automata, generalized urn models, and other contextual systems [1, 2, 3, 4]: pastings of Boolean algebras generated by partitions of a finite set, in which propositions within one context are jointly decidable while propositions from different contexts may be incompatible. A natural question is whether a *continuum* analogue—a pasting over a continuum state space—could approach or reproduce the propositional structure of quantum mechanics.

We answer this in two movements. *Negatively* (Section 2): the continuum extension via Boolean sub-σ-algebras of a common measurable space $(\Omega, \mathcal{F})$ is *concrete by construction*—point evaluations supply a full, order-determining family of two-valued states, instantiating the classical characterization of set-representable logics [5, 6]—so by the Kochen–Specker theorem [9] it cannot be isomorphic to $P(\mathcal{H})$ for $\dim \mathcal{H} \geq 3$, with Gleason's theorem [10] an independent route in the σ-additive setting. The obstruction is ontological: retaining a common point space is retaining a hidden-variable model.

Positively (Sections 3–4): the no-go is diagnostic. What the construction lacks is a *conservation law for probability across intertwined contexts*, which we decompose into four independent clauses:

(C1) *Non-contextuality of value*: a proposition shared by several contexts receives one value;

(C2) *Conservation in every context*: the values across each context sum to unity;

(C3) *Continuous transitive symmetry*: pure states form a connected manifold on which the reversible transformations act continuously and transitively;

(C4) *Budget rigidity*: the transition structure between pure states is itself the conserved quantity, and the symmetries are exactly those preserving it.

Each clause is enforced by a classical theorem—Kochen–Specker and Gleason for (C1)–(C2) at dimension ≥ 3, Hardy's continuity axiom for (C3), Wigner and Koecher–Vinberg for (C4)—and the clauses are logically independent (Section 4.2). Two observations frame the analysis. First, the qubit lattice $P(\mathbb{C}^2)$ is a horizontal sum, hence concrete, and Bell's 1966 model [11] reproduces its probabilities; at $d = 2$ the logical clauses do not bind, and the quantum content of the Bloch sphere—the equality $D^2 + V^2 = 1$ [13, 14, 15] and the Brukner–Zeilinger invariant [16, 17]—is carried by the geometric clauses alone (Section 4.3). Second, cardinality is not the missing ingredient: enlarging Ω enlarges a simplex but never makes it round, and concrete logics violate (C3) categorically (Lemma 3.2).

2 The Continuum Extension and Its Concreteness

Let Ω be a set equipped with a σ-algebra $\mathcal{F}$. We work with subsets of Ω directly (not modulo null sets), so that pointwise truth assignments are well defined.

Definition 2.1. A *context* on $(\Omega, \mathcal{F})$ is a Boolean sub-σ-algebra $\mathcal{C} \subseteq \mathcal{F}$. Given a family $\{\mathcal{C}_\alpha\}_{\alpha\in A}$ of contexts, the associated *concrete logic* is $L_\infty = \bigcup_{\alpha\in A} \mathcal{C}_\alpha$, partially ordered by inclusion, with $A^\perp = \Omega \setminus A$.

In the finite case each context is the algebra of unions of blocks of a partition, so this directly generalizes partition logic. We make no claim that L_∞ is orthomodular—unions of Boolean σ-algebras need not satisfy the orthomodular law, and ensuring it requires delicate pasting conditions [7]; our analysis needs only that L_∞ is a family of subsets of a common point space.

Example 2.2. Let $\Omega = S^2$ with its Borel σ-algebra. For each unit vector $\mathbf{n}$, let $\mathcal{C}_\mathbf{n}$ be generated by the partition of the sphere into open northern hemisphere, open southern hemisphere, and equator. Since $\mathcal{C}_{-\mathbf{n}} = \mathcal{C}_\mathbf{n}$, the contexts are indexed by $\mathbb{RP}^2$, and for $\mathbf{m} \neq \pm\mathbf{n}$ one has $\mathcal{C}_\mathbf{n} \cap \mathcal{C}_\mathbf{m} = \{\emptyset, \Omega\}$: a *horizontal sum*. We flag a coincidence for later use: the maximal contexts of the *qubit*, $\{P_\mathbf{n}, P_{-\mathbf{n}}\}$, are indexed by the same $\mathbb{RP}^2$. Example 2.2 is the *classicalization* of the qubit—same sphere, same context index, but with S^2 as *sample space* rather than *state space*.

For each $\omega \in \Omega$ define the *point evaluation* $v_\omega(A) = 1$ if $\omega \in A$, else 0. Each $v_\omega \colon L_\infty \to \{0, 1\}$ restricts to a Boolean homomorphism on every context: a *dispersion-free state*, sharp on every proposition in every context simultaneously.

Definition 2.3. A family $\mathcal{S}$ of two-valued states on a poset L is *order-determining* if $\big(\forall s \in \mathcal{S} : s(A) \le s(B)\big) \implies A \le B$.

Proposition 2.4. *$\{v_\omega\}_{\omega\in\Omega}$ is order-determining for L_∞.*

Proof. If $v_\omega(A) \le v_\omega(B)$ for all ω but $A \not\subseteq B$, pick $\omega_0 \in A\backslash B$; then $v_{\omega_0}(A) = 1 > 0 = v_{\omega_0}(B)$. □

This instantiates the classical characterization of concrete logics [5, 6]: a logic is set-representable iff it admits a full, order-determining set of two-valued states.

Corollary 2.5. *If $\dim \mathcal{H} \ge 3$, then L_∞ is not isomorphic to $P(\mathcal{H})$.*

Proof. Kochen–Specker [9] excludes any two-valued state on $P(\mathcal{H})$ for $\dim \mathcal{H} \ge 3$; L_∞ admits the v_ω. □

Remark 2.6 (Gleason). Independently, Gleason's theorem [10] shows that for $\dim \mathcal{H} \ge 3$ every countably additive probability measure on $P(\mathcal{H})$ is $\mathrm{Tr}(\rho\,\cdot)$; in particular no $\{0,1\}$-valued countably additive measure exists. The Kochen–Specker route is sharper, requiring no additivity beyond finite additivity within contexts.

The obstruction is thus ontological, not measure-theoretic: the points of Ω are hidden variables, and point evaluation is the deterministic valuation.

3 Dimension Two, and the Diagnosis

3.1 Where the obstruction does not apply

Proposition 3.1. *As an orthocomplemented poset, $P(\mathbb{C}^2)$ is the horizontal sum of continuum-many four-element Boolean algebras ($\mathrm{MO}_{\mathfrak{c}}$ in the notation of [8]) and admits a full, order-determining family of two-valued states; hence it is itself a concrete logic.*

Proof sketch. The nontrivial projections are rank one; two distinct ones commute iff complementary. The blocks $\{0, P, P^\perp, I\}$ intersect pairwise in $\{0, I\}$ and are indexed by $\mathbb{RP}^2$. A two-valued state may select, independently in each block, which of $P, P^\perp$ receives 1; additivity binds only within blocks, so all selections are admissible, and the family is order-determining. □

This is the order-theoretic face of Bell's observation [11] that the qubit admits a hidden-variable model reproducing all quantum probabilities: at dimension two the Kochen–Specker divide has not opened. Note that Example 2.2 and Proposition 3.1 exhibit horizontal sums over the *same* index set $\mathbb{RP}^2$: at $d = 2$ the classicalization of the qubit loses nothing *logically*; what it loses is metric—the transition probabilities between non-orthogonal states (Section 4.3). The structural moral: the obstruction of Corollary 2.5 is driven not by the multiplicity or cardinality of contexts but by their *intertwining*—the sharing of nontrivial propositions between maximal contexts, absent in any horizontal sum and dense in $P(\mathcal{H})$ for $\dim \mathcal{H} \ge 3$.

3.2 Cardinality is not the missing ingredient

The probability measures on $(\Omega, \mathcal{F})$ restrict to states on L_∞, forming a Choquet simplex whose extreme points are the v_ω; the reversible transformations compatible with the point structure are bijections of Ω, which merely *permute* the extreme points.

Lemma 3.2. *For any path $t \mapsto \omega(t)$ in Ω and $A \in L_\infty$, the function $t \mapsto v_{\omega(t)}(A) \in \{0,1\}$ is constant or discontinuous; and every v_ω is maximally sharp in every context simultaneously.*

Proof. Immediate: the function is $\{0,1\}$-valued, and v_ω restricts to a homomorphism on each $\mathcal{C}_\alpha$. □

Trivial as it is, Lemma 3.2 states precisely what a classical pasting can never do: realize a *continuous trade-off* in which sharpening the answers in one context continuously degrades those in another. Dispersion-free states have an unbounded information budget—they saturate all contexts at once—and the only reversible dynamics is permutation. Enlarging Ω from a finite set to a continuum enlarges the simplex; it does not bend it.

Example 2.2 now appears in its true light:

	Example 2.2	**Qubit (Bloch sphere)**
Role of S^2	sample space	pure-state space
A point is	a complete answer-book	a budget of potential answers
Rotations act on	points and contexts	states, conserving the budget
Sharpness	maximal in all contexts	maximal in one context only
Conserved quantity	none (vacuous)	$S_1^2 + S_2^2 + S_3^2 = 1$

The right-hand conserved quantity is the Brukner–Zeilinger operational information invariant [16], which for path/interference observables specializes to $D^2 + V^2 = 1$ [13, 14, 15]. But by Proposition 3.1, at dimension two this conservation law does not yet exclude dispersion-free valuations: it binds mixed states and continuous symmetries, not blockwise two-valued homomorphisms. The law acquires modal force—excluding things—only when contexts intertwine. The next section makes its anatomy explicit.

4 Principle C: Four Clauses of Contextual Conservation

Definition 4.1. Let L be a pasting of Boolean (σ-)algebras $\{\mathcal{C}_\alpha\}$, not necessarily concrete. A *probability budget* on L is a map $\mu\colon L \to [0,1]$ with $\mu(1) = 1$, (finitely or countably) additive within each context.

Noncontextuality is built in: μ is defined on propositions, not (proposition, context) pairs. On a horizontal sum the consistency requirement is vacuous; on a concrete logic it is satisfiable in $\{0,1\}$—by Proposition 2.4, lavishly so. The entire content of the quantum/classical divide is the behaviour of budgets under *intertwining*.

Principle C (Contextual conservation). A *state* of a contextual theory is a probability budget subject to:

- **(C1)** *Non-contextuality of value.* For every proposition $a \in L$ and all contexts $\mathcal{C}, \mathcal{C}'$ containing a, the value $\mu(a)$ is the same whether computed in $\mathcal{C}$ or in $\mathcal{C}'$.
- **(C2)** *Conservation in every context.* For every context $\mathcal{C}$ with atomic decomposition $\{a_1, a_2, \dots\}$, $\sum_i \mu(a_i) = 1$, with (countable) additivity within $\mathcal{C}$.
- **(C3)** *Continuous transitive symmetry.* The pure states form a connected topological manifold $\mathcal{P}$, and the group G of reversible transformations acts continuously and transitively on $\mathcal{P}$.

(C4) *Budget rigidity.* There is a continuous transition probability $p\colon \mathcal{P} \times \mathcal{P} \to [0,1]$, with $p(a,b) = 1$ iff $a = b$ and $p(a,b) = 0$ iff $a \perp b$, such that G is *exactly* the group of bijections of $\mathcal{P}$ preserving p; equivalently, the conserved invariant (purity; for the qubit, the squared Bloch length) determines the symmetry group, and conversely.

Clauses (C1)–(C2) are *logical* (the budget as a function on the pasted contexts); clauses (C3)–(C4) are *geometric* (the space of budgets and its symmetries). The dimension-two caveat is precisely the statement that the logical clauses can be satisfied classically at $d = 2$ while the geometric clauses already cannot.

4.1 Clause-wise enforcement

On $P(\mathcal{H})$ with $\dim \mathcal{H} \geq 3$ the rigidity of Principle C decomposes clause-wise:

1. **(C1)+(C2), two-valued: impossible** (Kochen–Specker [9]). A $\{0,1\}$-budget would have zero entropy in every context simultaneously; dense intertwining makes this combinatorially impossible. Contrapositively, a concrete logic is exactly one whose (C1)–(C2) constraints are slack enough to admit saturation everywhere (Section 2).

2. **(C1)+(C2),** $[0,1]$**-valued: unique** (Gleason [10]). Consistency across intertwined contexts alone forces $\mu = \mathrm{Tr}(\rho\,\cdot)$: the Born rule is the unique solution of the logical clauses in the σ-additive setting, with no assumption about dynamics, symmetry, or continuity.

3. **(C3): the classical/quantum separator** (Hardy [20]). In Hardy's reconstruction, classical probability and quantum theory differ in exactly one axiom: the existence of continuous reversible transformations between pure states—clause (C3). Lemma 3.2 shows no concrete logic can satisfy it: over a point space, pure states are vertices and reversible maps are permutations.

4. **(C4): the selector of complex quantum theory** (Wigner [12]; Koecher–Vinberg [24]). Wigner's theorem: the bijections of pure states preserving transition probabilities are exactly the (anti)unitaries, as (C4) demands. Homogeneity of the state cone (from (C3)) together with self-duality (the (C4) pairing of states with effects) characterizes the formally real Jordan algebras; adjoining local tomography selects complex quantum theory [20, 21, 22, 23]. That the complex case is empirically distinguishable has recently been demonstrated: real-Hilbert-space quantum theory is falsified in network Bell scenarios [25].

4.2 Independence: a scorecard

The clauses are logically independent; the proof is a museum of known theories, each deleting exactly one clause.

Theory	**(C1)**	**(C2)**	**(C3)**	**(C4)**
Classical probability; concrete logics over Ω	✓	✓	—	—
Spekkens' toy theory [19]	✓	✓	(discrete only)	—
Real-Hilbert-space quantum theory	✓	✓	✓	—
Jordan-algebraic state spaces (e.g. exceptional)	✓	✓	✓	—
Complex quantum theory, $\dim \mathcal{H} \geq 3$	✓	✓	✓	✓

Remark 4.2 (What each deletion costs). Deleting (C3) returns the simplex: pure states discrete, symmetries permutations, no interference—the entire family of partition logics and their continuum extensions (Lemma 3.2). Keeping only a discrete shadow of (C3) yields Spekkens' toy theory [19]: an epistemically restricted classical theory satisfying (C1)–(C2) that reproduces complementarity, no-cloning, and teleportation analogues—calibrating how much apparent quantumness resides in the logical clauses alone—but with no Bell violation, no continuous interference phase, no Gleason rigidity. Deleting (C4) while keeping (C3) yields the real and Jordan-algebraic alternatives, empirically distinguishable from the complex case and now excluded [25]. Deleting (C1) admits contextual value assignments and exits the framework; deleting (C2) severs the connection to operational probability. The conjunction carves out complex quantum theory, and no clause is redundant.

4.3 The qubit equation: four clauses in one formula

For the qubit, Principle C condenses into a single equation. For pure states ψ, ϕ at Bloch angle θ,

$$p(\psi, \phi) \;=\; \left|\langle\psi|\phi\rangle\right|^2 \;=\; \cos^2\frac{\theta}{2}. \tag{1}$$

Each clause is legible in (1): **(C1)** the value depends on the pair of states only; **(C2)** $\cos^2(\theta/2)+\sin^2(\theta/2) = 1$ in any basis $\{\phi, \phi^\perp\}$; **(C3)** θ varies continuously and $SO(3)$ acts transitively; **(C4)** $\cos^2(\theta/2)$ is, up to trivial reparametrization, the unique transition probability on S^2 admitting a transitive continuous invariance group of the Wigner type (cf. [18]).

Two consequences follow. First, $D^2+V^2 = 1$ and the Brukner–Zeilinger invariant are (1) read along orthogonal great circles: the (C3)–(C4) content of the qubit. Second, Proposition 3.1 shows the (C1)–(C2) content of the qubit to be classically satisfiable, and Example 2.2 shows its *logic* can be carried by a sample space; what no sample space can carry is (1) itself, since by Lemma 3.2 point evaluations support only the Kronecker structure $p(\omega, \omega') = \delta_{\omega\omega'}$. The quantum content of the Bloch sphere is not its logic but its *metric*. Textbook discussions of complementarity conducted entirely at $d = 2$ probe the conservation law only through its geometric clauses—which is precisely why it has so often been mistaken there for a triviality of normalization.

5 Conclusion: A Category Change, Not a Limit

The continuum extension of partition logics is faithful to the finite theory in every respect but one: it inherits, and cannot shed, the underlying point space. This single inheritance is fatal to any convergence toward quantum structure (Proposition 2.4, Corollary 2.5); the failure is invisible at dimension two (Proposition 3.1); and the correct diagnosis is the absence of a conservation law for probability across intertwined contexts, resolved here into four independent clauses. The logical clauses (C1)–(C2) are enforced at dimension ≥ 3 by Kochen–Specker and Gleason; the geometric clauses (C3)–(C4) by Hardy's continuity axiom and by Wigner's theorem with the Koecher–Vinberg classification and local tomography. Deleting (C3) yields classical probability and every partition logic; weakening it to a discrete action yields Spekkens' toy theory; deleting (C4) yields the real and Jordan-algebraic alternatives, now empirically excluded [25]. Their conjunction characterizes complex quantum theory. At dimension two the logical clauses are slack and the entire divide is carried by the metric (1), not the logic—which is why Bloch-sphere complementarity could be mistaken for a triviality of normalization.

A continuum limit of partition logics built on a single classical point space will always admit a complete answer-book behind the contexts, and a theory with an answer-book has nothing to conserve: its budgets saturate every context at once, its symmetries permute, and its transition structure is the Kronecker delta. The continuation of partition logics is therefore not a limit within the classical category but an exchange of its basic object: trade the hidden answer-book for the conserved budget. What remains of Bohr's celebrated analogy between complementarity and relativity is then exactly this—not the relativity of descriptions over a fixed stage, but a four-clause conservation law that holds precisely because no stage is left for it to play on.

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